



Emilio Gigante

Research assistant in robotics

EXPERIENCE

Project assistant

ACIN - TU Wien / AIT

Apr. 2026 - Present

Development of a Reinforcement Learning controller for the grasping task of an automated forestry crane (underactuated robotic arm).

EDUCATION

MSc in Robotics and Automation Engineering

University of Pisa

2022 - 2025

Thesis: Study of an approach for AUV missions managing through small language models (SLM) in order to allow on-board deployment, that includes failure handling and model confidence evaluation. The approach was applied to an AUV control system integrated with behavior trees.

BSc in Mechanical Engineering

University of Pisa

2017 - 2022

Thesis: Analysis of a gravitational accumulator for renewable energy storage.

PROJECTS

Navigation Project

University of Pisa

2024

Developed an integrated navigation system, that was used to estimate a short simulated trajectory of a marine vehicle. MATLAB and Simulink were used to generate the trajectory and to create simulated extended Kalman filters and navigation sensors as accelerometers, gyroscopes, magnetometers e UWB.

Guidance Project

University of Pisa

2024

Part of a team that entered and won a competition held in class, where we had to define the guidance law for a drone that had to intercept a simulated target in different scenarios inside a flying room. The law was implemented through Python.

Underwater Obstacle Avoidance

University of Pisa

2024

Part of a team that worked in a mission for an AUV that had to avoid simulated obstacles in a real lake. Our team goal was to process data from the sensors to update the map of the environment to allow the path planning.

Real-Time Systems Project

University of Pisa

2024

Developed a real-time 8-ball pool game with a graphic interface for one or two users to play a full game using the mouse and the keyboard. It was written in C and with a higher level library built using the p-thread library.

CODING

Python 4+ years

C/C++ 1 year

Git 2+ year

LaTeX 4+ years

TOOLS

MATLAB/Simulink 4+ years

ROS/ROS2 2+ years

Solidworks 2+ years

Mjlab/Mujoco 6 months

LANGUAGES

Italian native

English C1

German B1+

CONTACT

- Vienna, Austria
- emilio.gigante.eng@gmail.com
- Emilio Gigante
- Portfolio

OTHER INFO

- Driving license: Category B
- Basketball: competitive in high school, amateur now